





Final Thoughts

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## Surface Area

$$\begin{aligned} A &= 2\pi r^2 + 2\pi rh \\ &= 2\pi r(r+h) \\ &= 2\pi(3.43)(3.43+12.4) \\ &= 341 \text{ cm}^2 \end{aligned}$$

## Optimization

$$V = \pi r^2 \left( \frac{A}{2\pi r} - r \right)$$

$$V = \pi r^2 \left( \frac{341}{2\pi r} - r \right)$$

$$V' = 170.5 - 3\pi r^2$$

$$0 = 170.5 - 3\pi r^2$$

$$r = 4.25 \text{ cm}$$

$$h = \frac{A}{2\pi r} - r$$

## Volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi(3.43)^2(12.4) \\ &= 458 \text{ cm}^3 \end{aligned}$$

## Calculating h

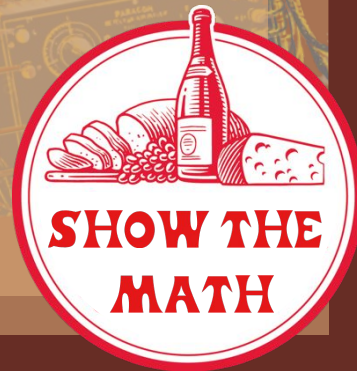
$$\begin{aligned} 341 \text{ cm}^2 &= 2\pi r(r+h) \\ &= 2\pi(4.25)(4.25+h) \\ h &= 8.52 \text{ cm} \end{aligned}$$

## Optimized Volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi(4.25)^2(8.52) \\ &= 483 \text{ cm}^3 \end{aligned}$$

## Percent Optimized

$$\begin{aligned} \%_O &= V_1/V_2 \times 100 \\ &= 458/483 \times 100 \\ &= 94.8\% \end{aligned}$$



~~Derivative~~

## Nutrition Facts

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$$V = \pi r^2 \left( \frac{341}{2\pi r} \right)$$

$$V' = \pi 2r \left( \frac{341}{2\pi r} - r \right) + \left( \frac{-341}{2\pi r^2} - 1 \right) \pi r^2$$

$$V' = 341 - 2\pi r^2 - 170.5 - \pi r^2$$

$$V' = 170.5 - 3\pi r^2$$

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## Surface Area

$$\begin{aligned}A &= 2\pi r^2 + 2\pi rh \\ &= 2\pi r(r+h) \\ &= 2\pi(2.86)(2.86+13.5) \\ &= 294 \text{ cm}^2\end{aligned}$$

## Optimization

$$\begin{aligned}V &= \pi r^2 \left( \frac{A}{2\pi r} - r \right) \\ V &= \pi r^2 \left( \frac{294}{2\pi r} - r \right)\end{aligned}$$

$$V' = 147 - 3\pi r^2$$

$$0 = 147 - 3\pi r^2$$

$$r = 3.95 \text{ cm}$$

$$h = \frac{A}{2\pi r} - r$$

## Volume

$$\begin{aligned}V &= \pi r^2 h \\ &= \pi(2.86)^2(13.5) \\ &= \pi(110.4) \\ &= 347 \text{ cm}^3\end{aligned}$$

## Calculating h

$$\begin{aligned}294 \text{ cm}^2 &= 2\pi r(r+h) \\ &= 2\pi(3.95)(3.95+h) \\ h &= 7.90 \text{ cm}\end{aligned}$$

## Optimized Volume

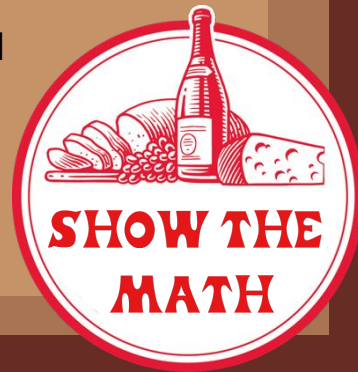
$$\begin{aligned}V &= \pi r^2 h \\ &= \pi(3.95)^2(7.90) \\ &= 387 \text{ cm}^3\end{aligned}$$

## Percent Optimized

$$\%_O = V_1/V_2 \times 100$$

$$= 347/387 \times 100$$

**89.7%**



~~Derivative~~

## Nutrition Facts

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$$V = \pi r^2 \left( \frac{294}{2\pi r} r \right)$$

$$V' = \pi 2r \left( \frac{294}{2\pi r} - r \right) + \left( \frac{-294}{2\pi r^2} - 1 \right) \pi r^2$$

$$V' = 294 - 2\pi r^2 - 147 - \pi r^2$$

$$V' = 147 - 3\pi r^2$$

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## Surface Area

$$\begin{aligned} A &= 2\pi r^2 + 2\pi rh \\ &= 2\pi(3.02)(3.02 + 15.5) \\ &= 351 \text{ cm}^2 \end{aligned}$$

## Optimization

$$V = \pi r^2 \left( \frac{A}{2\pi r} - r \right)$$

$$V = \pi r^2 \left( \frac{351}{2\pi r} - r \right)$$

$$V' = 175.5 - 3\pi r^2$$

$$0 = 175.5 - 3\pi r^2$$

$$r = 4.32 \text{ cm}$$

$$h = \frac{A}{2\pi r} - r$$

## Volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi (3.02)^2 (15.5) \\ &= \pi (141.361) \\ &= 444 \text{ cm}^3 \end{aligned}$$

## Calculating h

$$\begin{aligned} 351 \text{ cm}^2 &= 2\pi r(r+h) \\ &= 2\pi(4.32)(4.32+h) \\ h &= 8.61 \text{ cm} \end{aligned}$$

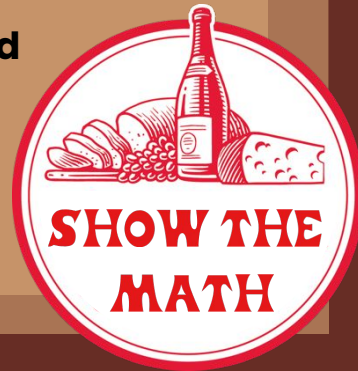
## Optimized Volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi(4.32)^2(8.61) \\ &= 505 \text{ cm}^3 \end{aligned}$$

## Percent Optimized

$$\begin{aligned} \%_O &= V_1/V_2 \times 100 \\ &= 444/505 \times 100 \end{aligned}$$

**87.9%**



~~Derivative~~

## Nutrition Facts

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$$V = \pi r^2 \left( \frac{351}{2\pi r} r \right)$$

$$V' = \pi 2r \left( \frac{351}{2\pi r} - r \right) + \left( \frac{-351}{2\pi r^2} - 1 \right) \pi r^2$$

$$V' = 351 - 2\pi r^2 - 175.5 - \pi r^2$$

$$V' = 175.5 - 3\pi r^2$$

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## Surface Area

$$\begin{aligned} A &= 2\pi r^2 + 2\pi r h \\ &= 2\pi(1.35)^2 + 2\pi(1.35)(8.5) \\ &= 83.6 \text{ cm}^2 \end{aligned}$$

## Optimization

$$V = \pi r^2 \left( \frac{A}{2\pi r} - r \right)$$

$$V = \pi r^2 \left( \frac{83.6}{2\pi r} - r \right)$$

$$V' = 41.8 - 3\pi r^2$$

$$0 = 41.8 - 3\pi r^2$$

$$r = 2.11 \text{ cm}$$

$$h = \frac{A}{2\pi r} - r$$

## Volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi(1.35)^2(8.5) \\ &= \pi(15.491) \\ &= 48.7 \text{ cm}^3 \end{aligned}$$

## Calculating h

$$\begin{aligned} 83.6 \text{ cm}^2 &= 2\pi r(r+h) \\ &= 2\pi(2.11)(2.11+h) \\ h &= 4.20 \text{ cm} \end{aligned}$$

## Optimized Volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi(2.11)^2(4.20) \\ &= 58.7 \text{ cm}^3 \end{aligned}$$

## Percent Optimized

$$\begin{aligned} \%_O &= V_1/V_2 \times 100 \\ &= 48.7/58.7 \times 100 \end{aligned}$$

**83.0%**



~~Derivative~~

## Nutrition Facts

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$$V = \pi r^2 \left( \frac{83.6}{2\pi r} r \right)$$

$$V' = \pi 2r \left( \frac{83.6}{2\pi r} - r \right) + \left( \frac{-83.6}{2\pi r^2} - 1 \right) \pi r^2$$

$$V' = 83.6 - 2\pi r^2 - 41.8 - \pi r^2$$

$$V' = 41.8 - 3\pi r^2$$

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## Surface Area

$$\begin{aligned}A &= 2\pi r^2 + 2\pi r h \\ &= 2\pi(3.50)(3.50 + 15.75) \\ &= 423 \text{ cm}^2\end{aligned}$$

## Optimization

$$V = \pi r^2 \left( \frac{A}{2\pi r} - r \right)$$

$$V = \pi r^2 \left( \frac{423}{2\pi r} - r \right)$$

$$V' = 211.5 - 3\pi r^2$$

$$0 = 211.5 - 3\pi r^2$$

$$r = 4.74 \text{ cm}$$

$$h = \frac{A}{2\pi r} - r$$

## Volume

$$\begin{aligned}V &= \pi r^2 h \\ &= \pi (3.50)^2 (15.75) \\ &= 606 \text{ cm}^3\end{aligned}$$

## Calculating h

$$\begin{aligned}423 \text{ cm}^2 &= 2\pi r(r+h) \\ &= 2\pi(4.74)(4.74+h) \\ h &= 9.46 \text{ cm}\end{aligned}$$

## Optimized Volume

$$\begin{aligned}V &= \pi r^2 h \\ &= \pi (4.74)^2 (9.46) \\ &= 667 \text{ cm}^3\end{aligned}$$

## Percent Optimized

$$\begin{aligned}\%_O &= V_1/V_2 \times 100 \\ &= 606/667 \times 100\end{aligned}$$

**90.9%**



~~Derivative~~

## Nutrition Facts

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$$V = \pi r^2 \left( \frac{423}{2\pi r} \right)$$

$$V' = \pi 2r \left( \frac{423}{2\pi r} - r \right) + \left( \frac{-423}{2\pi r^2} - 1 \right) \pi r^2$$

$$V' = 423 - 2\pi r^2 - 211.5 - \pi r^2$$

$$V' = 211.5 - 3\pi r^2$$

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# Final Thoughts

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We found that all of our cans were fairly well optimized (above 80%). The taller and more slender cans were less optimized. The optimization equation followed the rule:

$$\frac{1}{2}(\text{surface area}) - 3\pi r^2$$



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